

Time: 3Hours

Marks: 100

- N.B.: (1) All questions are compulsory.
 (2) Figures to the right indicate full marks.
 (3) Use of log table/ non-programmable calculator is allowed.

Q.1 Attempt ANY FOUR of the following:

- A) What is saponification? Explain its mechanism. 5
- B) List the characteristics of pericyclic reactions. Explain cycloaddition reactions with suitable examples. 5
- C) Distinguish between:
 i) acidity and electrophilicity 3
 ii) transition state and intermediate 2
- D) What does NGP stand for? Explain with a suitable example its effect on stereochemical outcome and kinetics of the reaction. 5
- E) Using a neat labelled Jablonski diagram, explain ISC and vibrational relaxation. 5
- F) What is a photochemical reaction? Explain di- π -methane reaction. 5

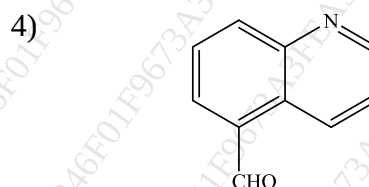
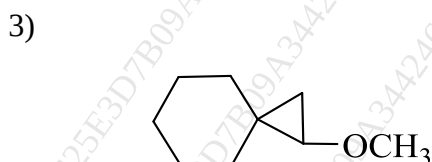
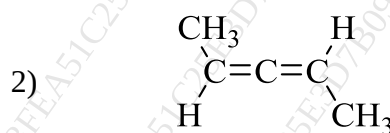
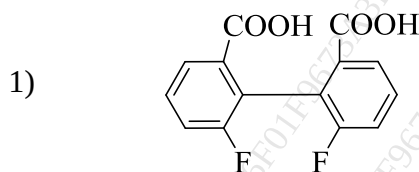
Q.2 Attempt ANY FOUR of the following:

- A) Write a note on the following- 5
 a) Plane of symmetry
 b) Centre of symmetry
- B) Explain the optical isomerism in cumulenes with even number of double bonds (Allenenes). 5
- C) Give the Skraup synthesis for the preparation of quinoline. Write the reaction of quinoline with the nitrating mixture. 5
- D) What is the action of following reagents on given heterocyclic compounds? 5
 a) $\text{H}_2\text{O}_2/\text{CH}_3\text{COOOH}$ on pyridine.
 b) SOCl_2 on pyridine-N-oxide.
 c) NaNH_2 in Liq.NH_3 on pyridine-N-oxide.
- E) What are Agrochemicals? Give two advantages of it. Write the synthesis of Indole-3-acetic acid. 5
- F) a) What are the properties of ideal pesticides? 3
 b) Draw the resonating structure of pyridine-N-Oxide 2

Q.3. Attempt ANY FOUR of the following:

- A) Write the structural formula for each of the following compounds: 5
- 1) Bicyclo[2.2.1] hept-2,5-diene
 - 2) Isoquinoline-1-carboxylic acid
 - 3) 1-methylspiro [3.3] heptane
 - 4) 2'-nitrodiphenyl-4-carboxylic acid
 - 5) 1-chloro-1,2-pentadiene

B) Give the IUPAC names of the following: **5**



C) a) Explain Linear synthesis with a suitable reaction? **3**

b) Give an example of regioselectivity? **2**

D) Give the synthesis of the following from a suitable starting compound? **5**

- 1) 2-methyl-2-propanol using a suitable Grignard reagent.
- 2) p-nitroaniline from aniline.

E) a) Calculate the percentage atom economy for the following reaction? **3**



b) Give the use of dimethyl carbonate as a selective methylating agent **2**

F) a) Explain multicomponent reaction with a suitable example? **3**

b) How is theoretical yield calculated? **2**

Q.4 Attempt ANY FOUR of the following:

A) a) Explain Chromophore theory in UV-visible spectroscopy **3**

b) Explain effect of solvent in UV-visible spectroscopy. **2**

B) a) Explain the mass spectral fragmentation pattern of ethyl methyl ketone **3**

b) Give the importance of molecular ion peak. **2**

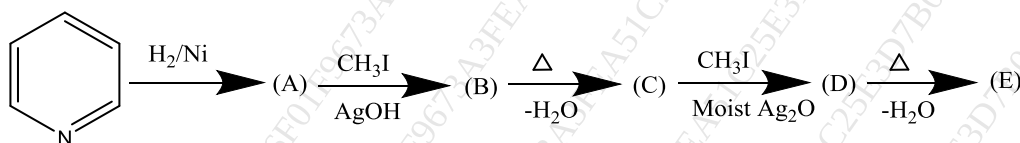
C) a) Give analytical evidence to prove the following:

1. Citral is an acyclic monoterpene **3**
2. The presence of an aldehydic group in Citral

b) What is Geminal dialkyl rule? **2**

D) Give the synthesis of Nicotine from Nicotinic acid. 5

E) a) Complete the following reaction 3



b) Write the structures of Geranial and Neral. 2

F) a) Give Ott's synthesis of adrenaline 3

b) What are the harmful effects of Nicotine? 2

Q.5 A) Select the correct option and complete the following statements: (ANY FIVE) 5

a) Thermal decomposition of having β -hydrogen to form alkenes is called Chugaev reaction.

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|---------------------|--------------------------|
| i) xanthate esters | ii) tertiary amine oxide |
| iii) tertiary amine | iv) acetates |

b) Acid catalysed esterification is areaction that follows..... mechanism.

- | | |
|-----------------------------|------------------------------|
| i) irreversible; $A_{Ac}2$ | ii) irreversible ; $A_{Ac}1$ |
| iii) reversible ; $A_{Ac}2$ | iv) reversible ; $A_{Ac}1$ |

c) The products of Cope elimination are

- | | |
|------------------------------|---|
| i) alkene + tertiary amine | ii) alkene + substituted hydroxyl amine |
| iii) alkane + tertiary amine | iv) alkane + substituted hydroxyl amine |

d) is an electron deficient species that attacks at positions of electron density

- | | |
|-----------------------|------------------------|
| i) Nucleophile; high | ii) Electrophile; high |
| iii) Nucleophile; low | iv) Electrophile; low |

e) reactions involve formation of two sigma bonds to the same carbon atom.

- | | |
|------------------|--------------------|
| i) Electrocyclic | ii) Group Transfer |
| iii) Sigmatropic | iv) Cheletropic |

f) Phosphorescence haswavelength and energy than incident radiation.

- | | |
|----------------------|--------------------|
| i) shorter; more | ii) longer; more |
| iii) shorter; lesser | iv) longer; lesser |

g) Photochemical conversion of benzophenone to benzpinacol is an example of

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|-----------------------|---------------------------------|
| i) photoisomerisation | ii) photochemical rearrangement |
| iii) photoreduction | iv) photochemical substitution |

h) Carbonyl compounds having γ -hydrogen under photolytic conditions prefer to undergo reaction.

- | | |
|------------------------------|---------------------|
| i) Norrish Type I | ii) Norrish Type II |
| iii) photochemical oxidation | iv) photoreduction |

- Q.5 B) State whether the following statements are TRUE or FALSE: (ANY FIVE) 5**
- Cumulenes with even number of double bonds and unsymmetrical substitution on terminal carbons will not show optical isomerism.
 - Always an optically active compound must contain at least one chiral carbon atom.
 - Centre of symmetry is also known as Inversion Centre.
 - In quinoline electrophilic substitution takes place at position 2.
 - Glycerol is reactant used in Skraup quinoline synthesis.
 - The dipole movement of pyridine-N-oxide is more than pyridine.
 - Fungicides are the chemicals that destroy, prevent or inhibit the growth of weeds.

- Q.5 C) Fill in the blanks with the correct alternative given in the bracket: (ANY FIVE) 5**
- [Atom economy, smaller, two, isolated, Roger. A.Sheldon, E-factor, fossil, green, Barry troast , cumulative, one, larger, catalytic]
- In Bicyclo [4.2.1] nonane, there exists _____ number of carbon atom between the two bridged carbon atoms
 - In 6-chlorospiro [4.5] decane , the substitution is on the _____ ring
 - Allene hascumulative double bonds.
 - Petroleum is an example of ----- fuel
 - reagents are superior to stoichiometric reagents
 - Reactions with lower ----- are desirable
 - The concept of atom economy was developed by -----
 - Liquid CO₂ is an example of a ----- solvent

- Q.5 D) Match the columns and rewrite the matched pairs: (ANY FIVE) 5**
- | Column A | Column B |
|--|------------------------------------|
| a) Citral | (i) Highest peak in mass spectrum |
| b) Adrenaline | (ii) Tobacco leaves |
| c) Nicotine | (iii) Stops blood haemorrhage |
| d) Base peak | (iv) Lemon grass oil |
| e) Hormones | (v) Hypsochromic shift |
| f) Isoprene | (vi) Monoterpenoid |
| g) Shift of absorption to lower wavelength | (vii) Ductless glands |
| | (viii) Bathochromic shift |
| | (ix) C ₅ H ₈ |
